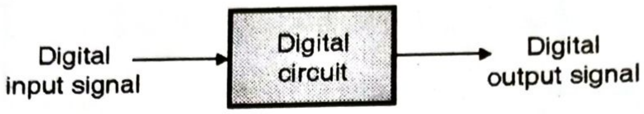
- The examples of analog systems are :

  1. Filters 2. Amplifiers 3. Signal generators

Digital Circuits (Systems) :

- These are the circuits that operate on the digital signals to produce digital signals.



(B-1706) **Fig. 1.3.1 : Digital circuit**

- The input signal to a digital system is digital and its output signal is also digital.

Examples of digital systems :

- The examples of digital circuits are adders, subtractors, registers, flip-flops, counters, microprocessors, digital calculators, computers etc.

1.4 Binary Logic and Logic Levels :

- A logic statement, is defined as a statement which is true if some condition is satisfied and false if that condition is not satisfied.
- For example, a bulb turns ON, if we close the switch, otherwise it is OFF.

1.4.1 Positive Logic :

SPPU : Oct. 04, April 18

University Questions

Q. 1 What is positive and negative logic ? Explain working of 2-input positive logic 'OR' gate using a diode circuit. **(Oct. 04)**

Q. 2 Explain the concept of '+ve logic' and '-ve logic' in case of logic gates. **(April 18, 4 Marks)**

- A "LOW" voltage level represents "logic 0" state and a comparatively "HIGH" output voltage level represents "logic 1" state, as shown in Fig. 1.4.1(a).
- For example, 0 Volts represent a logic 0 state and + 5V represent logic 1. This is called as "positive logic".

Positive Logic : Logic 0 (LOW) = 0V.

Logic 1 (HIGH) = + 5V.

1.4.2 Negative Logic :

SPPU : Oct. 04, April 18

University Questions

Q. 1 What is positive and negative logic ? Explain working of 2-input positive logic 'OR' gate using a diode circuit. **(Oct. 04)**

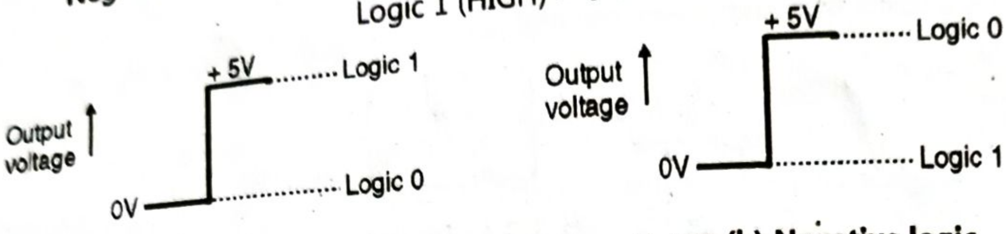
Q. 2 Explain the concept of '+ve logic' and '-ve logic' in case of logic gates. **(April 18, 4 Marks)**

- A "LOW" voltage level represents "logic 1" state and a "HIGH" output voltage level represents "logic 0" state, as shown in Fig. 1.4.1(b).

- For example, 0 Volts represent a "logic 1" state and + 5 V represent "logic 0" state. This is called as "negative logic".

Negative Logic : Logic 0 (LOW) = + 5 V

Logic 1 (HIGH) = 0 V.



(a) Positive logic

(B-440) (b) Negative logic

(B-439)**Fig. 1.4.1**

> **Note :** In this book, we are going to consider only the positive logic. Also we will assume the logic 0 level corresponds to 0 Volts and logic 1 level corresponds to + 5 V.

1.5 Number Systems :

Definition :

- A number system defines a set of values used to represent a quantity. We talk about the number of people attending class, the number of modules taken per student and also use numbers to represent grades obtained by students in tests.
- The study of number systems is not just limited to computers. We apply numbers every day and knowing how numbers work will give us an insight into how a computer manipulates and stores numbers.

1.5.1 Important Definitions :

SPPU : Oct. 07

University Questions	
Q. 1 What is meant by radix ?	(Oct. 07)

Radix or Base :

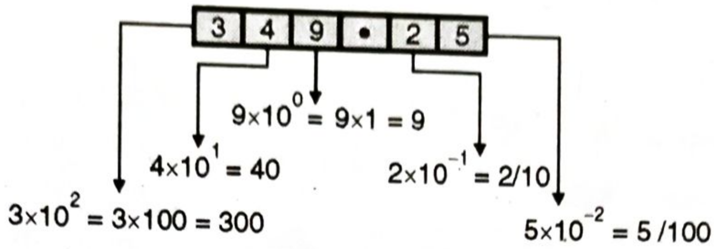
1. The number of values that a digit (one character) can have is equal to the base of the system. It is also called as the Radix of the system.

   For example for a decimal system, the base is "10" because every digit can have 10 distinct values. (0, 1, 2, ..., 9).

2. The largest value of a digit is always one less than the base :

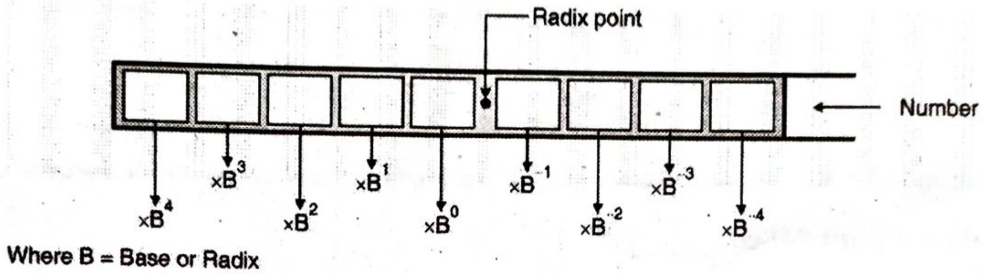
   For example, the largest digit in a decimal system is 9. (One less than the base 10).

3. **The position (place) of every digit represents a different multiple of base** i.e. the numbers have positional importance. For example consider the decimal number $(349.25)_{10}$ shown in Fig. 1.5.1.



(C-5) Fig. 1.5.1 : Numbers have positional importance

- Hence we can implement a common rule for all the numbering systems as follows. For a general number, we have to multiply each of digit by some power of **base (B) or radix** as shown in Fig. 1.5.2.



(C-6) Fig. 1.5.2

1.5.2 Various Number Systems :

SPPU : Oct. 10, April 13, April 14, April 17

University Questions

Q. 1 List different number systems. Give base (radix) of each. **(Oct. 10, 2 Marks)**

Q. 2 List any two number systems. Write their corresponding bases. **(April 13, 2 Marks)**

Q. 3 Give the radix of binary and hexadecimal number system. **(April 14, April 17, 2 Marks)**

Various number systems used in practice and their bases are as shown in Table 1.5.1.

Table 1.5.1 : Various number systems and their bases

Name of number system	Base
Binary	2
Octal	8
Decimal	10
Duodecimal	12
Hexadecimal	16

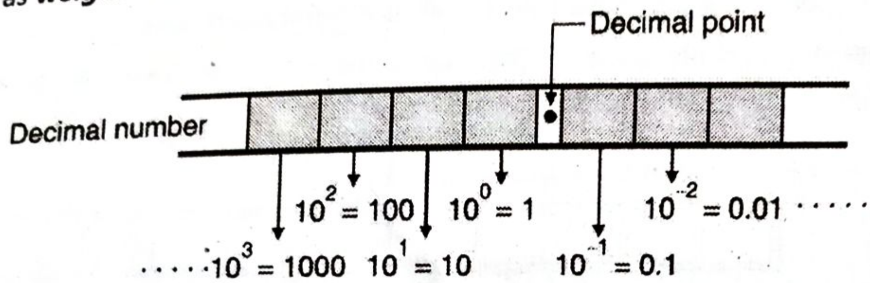
1.6 The Decimal Number System :

- The number system we have been taught to use since school is called the decimal numbering system.
- We are all familiar with counting and mathematics that uses this system.
- Looking at its make up will help us to understand other numbering systems.

1.6.1 Characteristics of a Decimal System :

- *Some of the important characteristics of a decimal system are :*

1. *It uses the **base of 10.***
2. *The largest value of a digit is 9.*
3. Each place (column number) represents a different multiple of 10. These multiples are also called as **weighted values**. The weighted values of each position are as shown in Fig. 1.6.1.



(C-8) Fig. 1.6.1 : Positions and corresponding weighted values for a decimal system

Most Significant Digit (MSD) :

The leftmost digit having the highest weight is called as the most significant digit of a number.

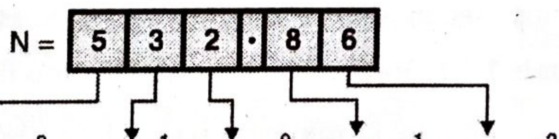
Least Significant Digit (LSD) :

The rightmost digit having the lowest weight is called as the least significant digit of a number.

Ex. 1.6.1 : Represent the decimal number 532.86 in terms of powers of 10.

Soln. :

The required representation is shown in Fig. P. 1.6.1.



$$N = 5 \times 10^2 + 3 \times 10^1 + 2 \times 10^0 + 8 \times 10^{-1} + 6 \times 10^{-2}$$

(C-9) Fig. P. 1.6.1

1.7 The Binary Number System :

SPPU : Oct. 07

University Questions

Q. 1 What is meant by LSB ? **(Oct. 07)**

- Most modern computer systems use the binary logic for their operation. A computer cannot operate on the decimal number system.
- A binary number system uses only two digits namely 0 and 1.
- The binary number system works like the decimal number system except one change. **It uses the base 2.**
- Hence the largest value of a digit is 1 and the number of values a digit can assume is two i.e. 0 and 1.
- The weighted values for different positions for a binary system are as shown in Fig. 1.7.1.

The Bit :

- The smallest "unit" of data is defined as a single **bit.** It can have two possible values i.e. 0 and 1.
- With a single bit we can represent any two distinct items like true or false, on or off, male or female, right or wrong etc. ,

The Nibble :

- A nibble is a combination of four bits. It would not be a particularly interesting data structure except for two items : BCD (Binary Coded Decimal) numbers and hexadecimal (base 16) numbers because it takes four bits to represent a single BCD or hexadecimal digit.
- With a nibble, we can represent up to 16 distinct values i.e. from 0000 to 1111.

  That means,

  Number of distinct values = $2^N = 2^4 = 16$

- where, N=Number of bits used to represent a number.

The Byte :

A byte is a combination of 8-binary bits. It consists of two nibbles.The number of distinct values represented by a byte is 256, ranging from 00000000 to 11111111.

The Word :

A word is a combination of 16 bits.Hence it consists of two bytes.

1.8 Hexadecimal Number System :

SPPU : Oct. 05

University Questions

Q. 1 Describe hexadecimal number system in detail. How can decimal numbers be converted to hexadecimal numbers ? **(Oct. 05)**

- The important features of a hexadecimal number system are as follows :
- **Base** : The base of hexadecimal system is 16.
- **Number of values assumed by each digit :** The number of values assumed by each digit is 16. The values include digits 0 through 9 and letters A, B, C, D, E, F. Hence the sixteen possible values are : 0 1 2 3 4 5 6 7 8 9 A B C D E F
- 0 represents the least significant digit whereas F represents the most significant digit.
- The base of 16 needs 16 digits. Hence it borrows 0 through 9 but needs another 6. For these the first six letters A, B, C, D, E, F are used. Here A represents 10, B represents 11 and so on. The hexadecimal digits and their values are as shown in Table 1.8.1.

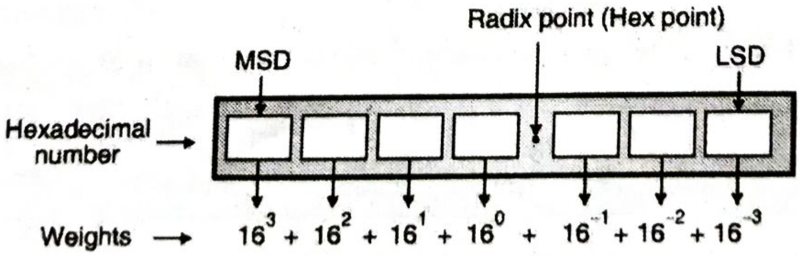
Table 1.8.1 : Hexadecimal digits and their values

Hexadecimal digit	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Value	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Table 1.8.2 : Hexadecimal numbers

0	10	20	30	....	90	A0	....	F0
1	11	21	31	....	91	A1	....	F1
2	12	22	32	....	92	A2	....	F2
3	13	23	33	....	93	A3	....	F3
4	14	24	34	....	94	A4	....	F4
.	.	.	.		.	.		.
.	.	.	.		.	.		.
.	.	.	.		.	.		.
.	.	.	.		.	.		.
D	1D	2D	3D	....	9D	AD	....	FD
E	1E	2E	3E	....	9E	AE	....	FE
F	1F	2F	3F	....	9F	AF	....	FF

- When dealing with large values, binary numbers quickly become too unwidely. The hexadecimal (base 16) numbering system solves this problem. Hexadecimal numbers offer the two features :
  1. Hex numbers are very compact.
  2. It is easy to convert from hex to binary and binary to hex.
- **Largest value of a digit :** The largest value of a digit in the hexadecimal number system is 15 and it is represented by F. The hexadecimal number higher than F will be 10. Table 1.8.2 gives you a clear idea about hexadecimal numbers.
- The largest two digit hexadecimal number is FF which corresponds to 255 decimal. The next higher number after FF is 100.
- **Positional weights :** The positional weights for a hexadecimal number to the left and right of decimal point are as shown in Fig. 1.8.1.

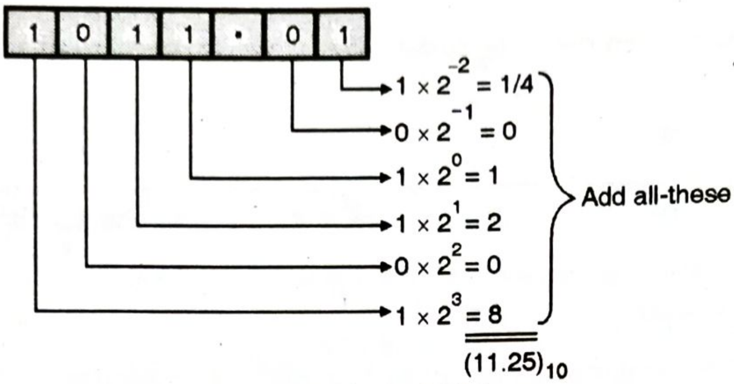


(C-15) Fig. 1.8.1 : Positional weights for a hexadecimal system

Ex. 1.8.1 : Represent the hexadecimal number 6DE in the powers of 16.

Soln. :

Steps 1, 2 and 3 :



(C-19) Fig. P. 1.10.1

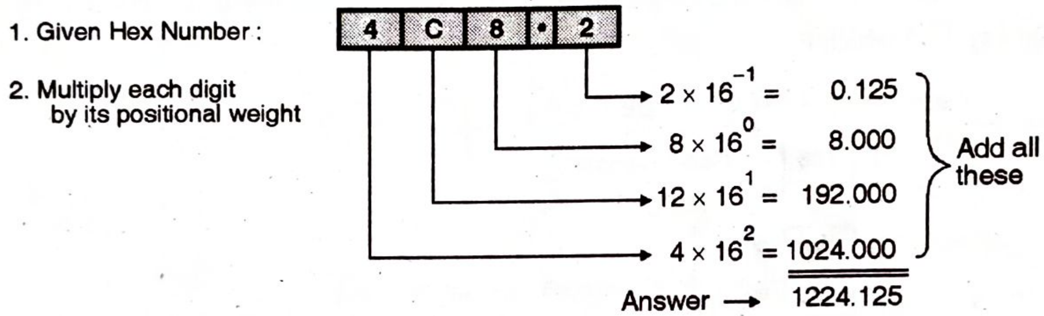
Step 4 : Addition :

$$\therefore \quad (1011.01)_2 = (11.25)_{10}$$ **...Ans.**

Hex to Decimal Conversion :

Ex. 1.10.2 : Convert the hex number $(4C8.2)_{16}$ into its equivalent decimal number.

Soln. :



$$\therefore (4C8.2)_{16} = (1224.125)_{10}$$

(C-21) Fig. P. 1.10.2

1.10.2 Conversion from Decimal to Other Systems :

SPPU : Oct. 05

University Questions

Q. 1 Describe hexadecimal number system in detail. How can decimal numbers be converted to hexadecimal numbers ? **(Oct. 05)**

- Decimal to binary conversion is also known as **Double-Dabble method** and Decimal to hexadecimal conversion is known as **Hex-Dabble method.**
- If the given decimal number consists of a decimal point, then we have to first separate out the integer and fractional parts.
- Then convert them separately to the desired radix, and combine the converted parts to obtain the complete converted number.
- The procedures for converting the integer part and the fractional part are completely different from each other.

1.11 Conversion from Binary to Other Systems :

1.11.1 Conversion from Binary to Decimal :

We have already discussed this.

1.11.2 Binary to Hex Conversion :

It is easy to convert from an integer binary number to hex. This is accomplished by following the procedure given bellow :

Step 1 : Divide the binary number into 4-bit sections from the LSB to the MSB.

Step 2 : Convert each 4-bit binary number to its hex equivalent.

Ex. 1.11.1 : Convert the binary number 1010 1111 1011 0010 into the equivalent hex number.

Soln. :

1.	Split the given number into group of 4 bits.	**1010**	**1111**	**1011**	**0010**
2.	Convert each group into corresponding hex.	A	F	B	2

(C-36(a))

$\therefore \quad (1010\ 1111\ 1011\ 0010)_2 = (AFB2)_{16}$ ...Ans.

1.12 Conversion from Other Systems to Binary System :

1.12.1 Conversion from Decimal to Binary :

We have already discussed this earlier.

1.12.2 Hex to Binary Conversion :

It is also easy to convert from an integer hex number to binary. This is accomplished by :

Step 1 : Convert each hex digit to its 4-bit binary equivalent.

Step 2 : Combine the 4-bit sections by removing the spaces.

Ex. 1.12.1 : Convert the hex number AFB2 into equivalent binary number.

Soln. :

Each digit in the given hex number is converted into 4-bit binary numbers as shown in Fig. P. 1.12.1.

Given hex number	A	F	B	2
Each digit converted to its four bit binary equivalent	1010	1111	1011	0010

(C-38) Fig. P. 1.12.1 : Hex to binary conversion

Hence $(A\ F\ B\ 2)_{16} = (1010\ 1111\ 1011\ 0010)_2$ **...Ans.**

1.13 Conversions Related to Hexadecimal System :

1.13.1 Other Systems to Hex :

– We are supposed to discuss the following conversions :

1. Decimal to Hex
2. Binary to Hex

We have already discussed them.

1.13.2 Hex to Other Systems :

– We are supposed to discuss the following conversions :

1. Hex to Decimal
2. Hex to Binary

– We have already discussed them.

1.14 Solved University Examples on Number Systems :

Ex. 1.14.1 : Perform : $(23.85)_{10} = (?)_2$ **April 12, 2 Marks**

Soln. :

Step 1 : Separate integer and fractional part :

$$\underbrace{23}_{\text{Integer}} \cdot \underbrace{85}_{\text{Fractional part}}$$

(C-4455)

2.1 Concept of Coding :

- When numbers, letters or words are represented by a specific group of symbols, it is said that the number, letter or word is being **encoded.**
- And the group of symbols is called as the **code.**
- The digital data is represented, stored and transmitted as group of binary bits. This group is also called as **binary code.**
- The binary codes can be used for representing the numbers as well as alphanumeric letters.

2.2 Classification of Codes :

2.2.1 Weighted Binary Codes :

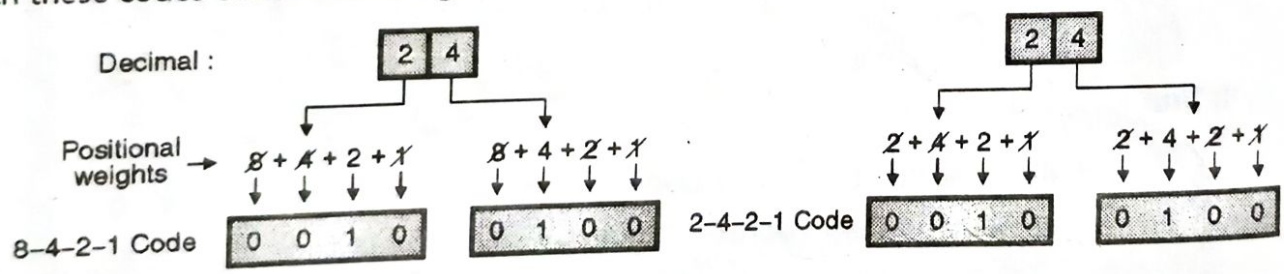
SPPU : April 08, April 12, Oct. 17

University Questions

Q. 1	Differentiate between : Weighted and Unweighted codes.	(April 08, Oct. 17, 2 Marks)
Q. 2	What is weighted number system ? Give any example of such a system.	(April 12, 2 Marks)

Definition :

- Weighted binary codes are those which obey the positional weight principle.
- Each position of the number represents a specific weight.
- Several systems of codes are used to express the decimal digits 0 through 9. 8421, 2421, 3321 .... all are the weighted codes.
- In these codes each decimal digit is represented by a group of four bits as shown in Fig. 2.2.1.



(C-73) Fig. 2.2.1

2.2.2 Non Weighted Codes :

SPPU : April 08, Oct. 17

University Questions

Q. 1	Differentiate between : Weighted and Unweighted codes.	(April 08, Oct. 17, 2 Marks)

Definition :

- In these codes, the positional weights are not assigned are known as non weighted codes.
- The examples of non-weighted codes are excess-3 and gray codes.

2.2.3 Alphanumeric Codes :

Oct. 17

University Questions

...n alphanumeric codes. (Oct. 17, 4 Marks)

- The special code designed to represent numbers as well as alphabetic characters are called as the alphanumeric codes.
- Some of these codes are capable to representing some symbols and instructions as well, in addition to the numbers and alphabetic characters.
- Examples of alphanumeric codes are : ASCII (American Standard Code for Information Interchange), EBCDIC (Extended Binary Coded Decimal Interchange Code) and Hollerith code.

2.3 Binary Coded Decimal (BCD) Code : SPPU : Oct. 11

University Questions

Q. 1 What is BCD code ? Convert decimal number $(14)_{10}$ BCD and to binary. **(Oct. 11, 4 Marks)**

- BCD is the short form of "Binary Coded Decimal." In this code each decimal digit is represented by a 4-bit binary number.
- Thus BCD is a way to express each of the decimal digits with a binary code.
- The positional weights associated to the binary bits in BCD code are 8-4-2-1 with 1 corresponding to LSB and 8 corresponding to MSB. These weights are actually 2^3, 2^2, 2^1 and 2^0 which are same as those used in the normal binary system.

Conversion from decimal to BCD :

- The decimal digits 0 to 9 are converted into a BCD, exactly in the same way as binary .
- For example decimal 4 corresponds to 0100 BCD which is same as binary. Similarly the BCD numbers corresponding to the decimal numbers 0 to 9 are exactly same as the corresponding binary numbers. This is illustrated in Table 2.3.1.

(C-6142) Table 2.3.1

Decimal	0	1	2	3	4	5	6	7	8	9
BCD	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001

BCD numbers are same as 4 bit binary numbers

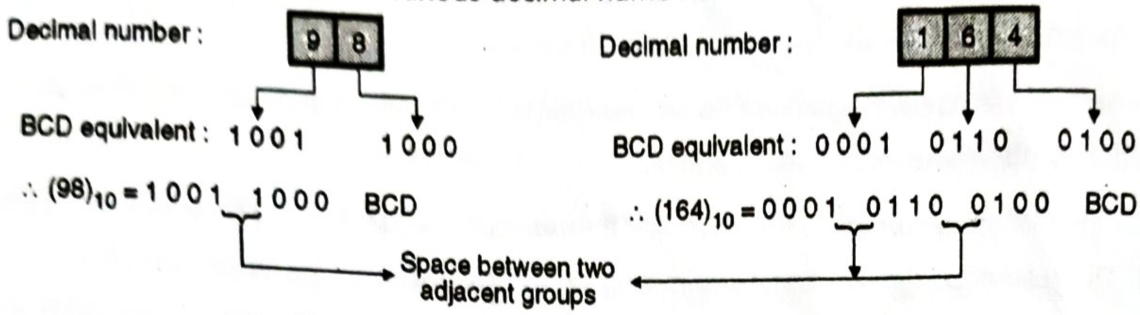
Invalid BCD codes :

- With four bits we can represent sixteen numbers (0000 to 1111). But in BCD code only the first ten of these are used (0000 to 1001).
- The remaining six code combinations i.e. 1010 to 1111 are invalid in BCD.

Conversion of big decimal numbers to BCD :

- To express any decimal number to BCD, simply express each decimal digit with its equivalent 4-bit BCD code as illustrated below.

- Fig. 2.3.1 shows the BCD codes for various decimal numbers.



(C-74) **Fig. 2.3.1 : Decimal to 8-4-2-1 BCD conversion**

Smallest and largest number in BCD :

The smallest number in BCD is (0000) and the largest one is (1001) i.e. 9. The next number to 9 will be $(10)_{10}$ which is expressed as (0001 0000) in BCD.

Advantages of BCD codes :

1. It is very similar to decimal system.
2. We need to remember binary equivalents of decimal numbers 0 to 9 only.

Disadvantages :

1. The addition and subtraction of BCD have different rules.
2. The BCD arithmetic is little more complicated.
3. BCD needs more number of bits than binary to represent the same decimal number. So BCD is less efficient than binary.

Ex. 2.3.1 : 1101 is not a BCD number. Justify. **April 04**

Soln. :

The largest BCD four bit number is 1001 i.e. $(9)_{10}$. So 1101 which is $(13)_{10}$ is an invalid BCD number.

2.3.1 Comparison with Binary :

SPPU : Oct. 06, Oct. 11

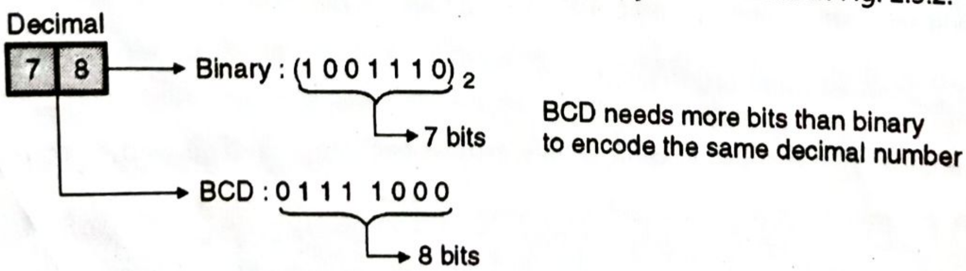
University Questions

Q. 1 What is the difference between BCD and binary code ? Give the BCD and binary equivalent of $(15)_{10}$. **(Oct. 06)**

Q. 2 What is BCD code ? Convert decimal number $(14)_{10}$ BCD and to binary. **(Oct. 11, 4 Marks)**

1. **BCD is less efficient than binary :**

Conversion of a decimal number 78 into BCD and binary is illustrated in Fig. 2.3.2.



(C-75) **Fig. 2.3.2 : BCD is less efficient than binary**

Fig. 2.3.2 illustrates that to encode the same decimal number, BCD needs more number of bits than binary. Hence BCD is less efficient as compared to binary.

2. BCD arithmetic is more complicated than Binary arithmetic.
3. The advantage of a BCD code is that the conversion from Decimal to BCD or vice versa is simpler.
4. Table 2.3.2 shows the comparison of binary and BCD numbers from 0 to 15 decimal.

(C-6143)Table 2.3.2

Decimal	Binary	BCD
0	0000	0000
1	0001	0001
2	0010	0010
3	0011	0011
4	0100	0100
5	0101	0101
6	0110	0110
7	0111	0111
8	1000	1000
9	1001	1001
10	1010	0001 0000
11	1011	0001 0001
12	1100	0001 0010
13	1101	0001 0011
14	1110	0001 0100
15	1111	0001 0101

BCD and binary numbers are same (0 to 9)

BCD and binary numbers are different (10 to 15)

What is meant by packed BCD ?

The BCD numbers corresponding to decimal numbers beyond 9 are called as packed BCD. Some examples of packed BCD are presented in Table 2.3.3.

Table 2.3.3 : Examples of packed BCD

Decimal	Packed BCD		
25		0010	0101
169	0001	0110	1001
523	0101	0010	0011

Ex. 2.3.2 : Convert the following decimal numbers to BCD.

(a) 35 (b) 174 (c) 2479

Soln. :

	(a)		(b)			(c)			
Decimal	3	5	1	7	4	2	4	7	9
BCD	0011	0101	0001	0111	0100	0010	0100	0111	1001

(C-6143(a))

Ex. 2.4.2 : Convert $(396)_{10} = (?)_{Excess\ 3}$ **April 12, 4 Marks**

Soln. :

$(396)_{10} = (?)_{Excess-3}$:

Decimal number :	3	9	6
	↓	↓	↓
Excess–3 :	0110	1100	1001

$\therefore\ (396)_{10} = (011011001001)_{Excess-3}$ ...**Ans.**

2.5 Gray Code :

SPPU : Oct. 04, Oct. 05, Oct. 07

University Questions

Q. 1	Gray code is not used for performing arithmetic operations, why ?	(Oct. 04)
Q. 2	Write a short note on gray code.	(Oct. 05)
Q. 3	What is gray code ? Explain how to convert a binary number to gray code.	(Oct. 07)

- Gray code is another non-weighted code. It is not an arithmetic code.
- It has a very special feature that only one bit in the gray code will change, each time the decimal number is incremented as shown in Table 2.5.1.
- As only one bit changes at a time, the Gray code is called as a **unit distance code.** The Gray code is a **Cyclic code.**

(C-91) Table 2.5.1

Decimal	Binary $B_3\ B_2\ B_1\ B_0$	Gray $G_3\ G_2\ G_1\ G_0$
0	0 0 0 0	0 0 0 0
1	0 0 0 1	0 0 0 1
2	0 0 1 0	0 0 1 1
3	0 0 1 1	0 0 1 0
4	0 1 0 0	0 1 1 0
5	0 1 0 1	0 1 1 1
6	0 1 1 0	0 1 0 1
7	0 1 1 1	0 1 0 0
8	1 0 0 0	1 1 0 0
9	1 0 0 1	1 1 0 1
10	1 0 1 0	1 1 1 1
11	1 0 1 1	1 1 1 0
12	1 1 0 0	1 0 1 0
13	1 1 0 1	1 0 1 1
14	1 1 1 0	1 0 0 1
15	1 1 1 1	1 0 0 0
16	1 0 0 0 0	0 0 0 0

$G_3 = B_3$
$G_2 = B_3 \oplus B_2$
$G_1 = B_2 \oplus B_1$
$G_0 = B_1 \oplus B_0$

Note :

1. In gray code only the encircled bit changes when the decimal number is incremented by 1.
2. In binary one, two, three or sometimes all the 4 bits change when the decimal number is incremented by 1.

- The gray code possesses another important property. It states that the gray code for a decimal number ($2^n - 1$) where n = 2, 3, 4 ..., will differ from gray 0 (0000) in only 1 bit position. This is illustrated in Fig. 2.5.1.

n	Decimal number $2^n - 1$	Gray code
2	$2^2 - 1 = (3)_{10}$	0 0 1 0
3	$2^3 - 1 = (7)_{10}$	0 1 0 0

This gray code differs from gray 0(0000) only in this position

(C-92) Fig. 2.5.1

2.5.1 Applications of Gray Code :

- Gray code is popularly used in the shaft position encoders.
- A shaft position encoder produces a code word which represents the angular position of the shaft.

$\therefore \quad (1001)_2 = (1101)_{Gray}$ ...**Ans.**

2.6 Alphanumeric Codes :

- A binary digit or bit can represent only two symbols as it has only two states "0" or "1". But this is not enough for communication between two computers because there we need many more symbols for communication. These symbols are required to represent :
- 26 alphabets with capital and small letters.
- Numbers from 0 to 9.
- Punctuation marks and other symbols.
- **Alphanumeric codes** are the codes that represent numbers and alphabetic characters (letters). Mostly such codes also represent other characters such as symbol and various instructions necessary for conveying information.
- An alphanumeric code should at least represent 10 decimal digits and 26 letters of alphabet i.e. total 36 items.
- Therefore instead of using only single binary bits, a group of bits is used as a code to represent a symbol. The number of bits used per code word will be dependent on the total number of symbols to be represented e.g. if 5 bits are used per code word then $2^5 = 32$ combination would be possible. If the word length is increased to 8 then the number of combinations would be $2^8 = 256$, hence an 8-bit code can represent 256 symbols.

- The following three alphanumeric codes are very commonly used for the data representation :
  1. American Standard Code for Information Interchange (ASCII)
  2. Extended Binary-Coded-Decimal Interchange Code (EBCDIC)
  3. Five bit Baudot code.
- ASCII code is a 7-bit code whereas EBCDIC is an 8-bit code. ASCII code is more commonly used worldwide while EBCDIC is used primarily in large IBM computers.

2.6.1 ASCII - (American Standard Code for Information Interchange) :

SPPU : Oct. 05, April 06, April 16

University Questions

Q. 1 What is ASCII code ? Where is it used ? **(Oct. 05, April 06)**

Q. 2 Write the full form of ASCII. **(April 16, 2 Marks)**

- ASCII is the abbreviation of American Standard Code for Information Exchange. It is a universally accepted alphanumeric (character) code.
- It is used in most computers and other electronic equipment. Most computer keyboards are standardized with ASCII. When we press a key, the corresponding ASCII code is generated which goes into the computer.
- ASCII has 128 characters and symbols. We need 7 bits to represent 128 characters. So ASCII is a 7 bit code.
- It can be considered as an 8 bit code with MSB = 0 always.
- This 8-bit code is 00 through 7F in Hexadecimal.
- The first thirty-two ASCII characters are nongraphic commands which are never printed or displayed. They are used only for the control purpose.
- Examples of control characters are "line feed" or "null" or "escape" etc.
- The remaining characters are graphic symbols which can be displayed or printed. This includes the letters of alphabets, ten decimal digits, punctuation signs and other commonly used symbols.
- The ASCII code set consists of 94 printable characters, SPACE and DELETE characters, and 32 control symbols. This is as shown in Table 2.6.1. Two examples of ASCII are as given below :

$$A = 41 = 1000001$$

$$B = 42 = 1000010$$

Sr. No.	BCD	ASCII
1.	It is a weighted binary code.	It is an alphanumeric code
2.	It uses 4 bits to represent a decimal digit.	It uses 7 or 8 bits to represent a character or a number.
3.	It can represent only numbers, not the characters.	It can represent numbers, symbols as well as characters.
4.	It can be used for internal arithmetic manipulations.	It is used for communication between computers, peripherals etc.

2.7 Solved University Examples on Codes :

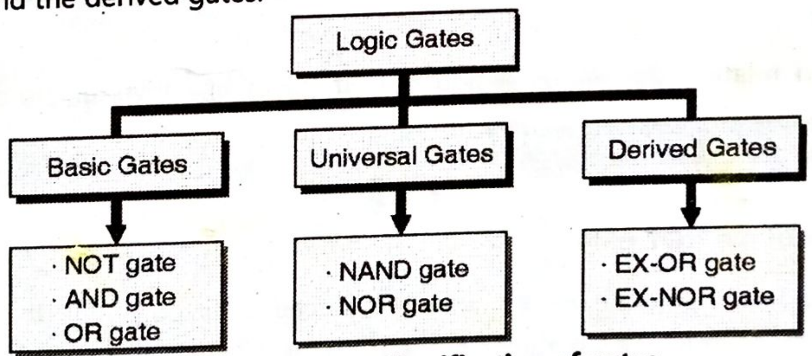
3.2 Logic Gates :

Definition :

- Logic gates are the logic circuits which act as the basic building blocks of any digital system. It is an electronic circuit having one or more than one inputs and only one output.
- The relationship between the input and the output is based on a "certain logic". Depending on this logic, the gates are named as NOT gate, AND gate, OR, NAND, NOR etc.
- We have to use different laws, rules and theorems to analyze the digital circuits.
- By connecting the gates mentioned earlier, in different ways, we can build circuits that can perform arithmetic and other functions associated with the human brain.
- Because they simulate mental processes, gates are often called as logic circuits.

3.2.1 Classification of Logic Gates :

As shown in Fig. 3.2.1, the logic gates are classified into three categories namely, the basic gates, the universal gates and the derived gates.



(B-448) **Fig. 3.2.1 : Classification of gates**

3.2.2 Truth Table :

SPPU : Oct. 07

University Questions

Q. 1 What is truth table ? **(Oct. 07)**

The operation of a logic gate or a logic circuit can be best understood with the help of a table called "Truth Table". The truth table consists of all the possible combinations of the inputs and the corresponding state of output produced by that logic gate or logic circuit.

3.2.3 Boolean Expression :

- The relation between the inputs and the outputs of a gate can be expressed mathematically by means of the **Boolean Expression**. Let us now discuss the operation of various logic gates.

3.3 NOT Gate or Inverter :

SPPU : April 14, Oct. 14, Oct. 16

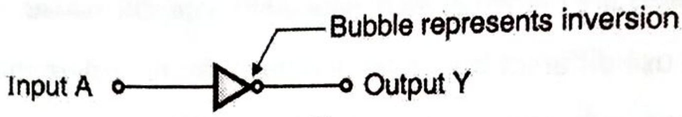
University Questions

Q. 1	Draw the symbol and truth table of NOT and three input XOR gates.	(April 14, 4 Marks)
Q. 2	Draw logic symbol and truth table of NOR gate and NOT gate.	(Oct. 14, 2 Marks)
Q. 3	Write truth table of 2 input AND gate and NOT gate.	(Oct. 16, 2 Marks)

Logical symbol and Truth Table :

The NOT gate or Inverter is a logic gate having one input (A) and one output (Y). Its symbol and truth table are shown in Fig. 3.3.1.

Symbol :



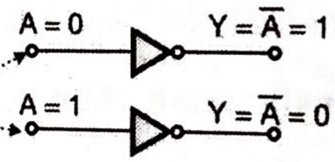
Boolean equation : $Y = \bar{A}$

(a)

Truth table

Input A	Output Y
0	1
1	0

(b)



(c) Equivalent circuits

(B-450) **Fig. 3.3.1 : Symbol, truth table and equivalent circuit of a NOT gate**

Boolean expression :

The expression relating the inputs A and output (Y) of an inverter is called as its Boolean expression. The Boolean expression for an inverter is,

$$Y = \bar{A}$$

Important points about the NOT gate :

1. The NOT gate is also known as an "Inverter" because its output is the inverted version or "complement" of its input. This is shown in the truth table of NOT gate.
2. The bubble (o) in the symbol of a NOT gate indicates the inversion operation.
3. The operation of a NOT gate for various input combinations is as shown in Fig. 3.3.1(c).

3.4 AND Gate :

SPPU : Oct. 15, Oct. 16, Oct. 17, April 18

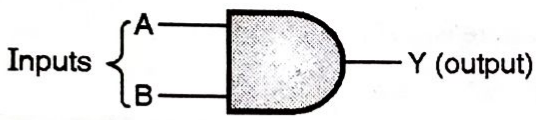
University Questions

Q. 1	Draw logic symbol and write truth table of two input AND and EXOR gates.	(Oct. 15, 2 Marks)
Q. 2	Write truth table of 2 input AND gate and NOT gate.	(Oct. 16, 2 Marks)
Q. 3	Draw the symbol and give the truth table for : 1. 2-input AND gate 2. 2-input OR gate	(Oct. 17, April 18, 4 Marks)

- AND is one of the logic operators. It performs the logical multiplication on its inputs.
- The output is high (Y = 1) if and only if all the inputs to the AND gate are high (1). The output is low (0), if any one or more inputs are low (0). AND gate can have two or more inputs and only one output.

Logical symbol and Truth Table :

The logical symbol of a two input AND gate is as shown in Fig. 3.4.1(a). The truth table of a two input AND gate is shown in Fig. 3.4.1(b). Note that the output is HIGH (1) only when both the inputs are HIGH (1).



(a) Logical symbol

Inputs		Output
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(b) Truth table

(B-453) **Fig. 3.4.1**

Boolean expression :

The expression relating the inputs (A, B) and output (Y) of a gate is called as the Boolean expression. The Boolean expression for an AND gate is,

$$Y = A \cdot B$$

where the "dot" between A and B represents multiplication.

3.5 The OR Gate :

SPPU : April 15, Oct. 16, Oct. 17, April 18

University Questions

Q. 1 Draw the symbol and truth table for 2 input OR gate and 2 input NAND gate. **(April 15, 2 Marks)**

Q. 2 Draw circuit diagram for 2 input OR gate using discrete components. Explain its working. **(Oct. 16, 4 Marks)**

Q. 3 Draw the symbol and give the truth table for :
1. 2-input AND gate 2. 2-input OR gate **(Oct. 17, April 18, 4 Marks)**

- An "OR" gate performs the logical addition on its inputs therefore its output will be high (1) if any one or both the inputs are high (1).
- Its output will be low (0) if and only if both the inputs are simultaneously low (0).

Logical symbol :

Fig. 3.5.1(a) shows the logical symbol of a two input OR gate.

Truth table :

- Fig. 3.5.1(b) shows the truth table for a two input OR gate. Note that the output Y is LOW (0) if and only if both the inputs are LOW (0).
- If any one or both the inputs are HIGH (1), then the output Y is HIGH (1).

(a) Logical symbol

Inputs		Output
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(b) Truth table

(B-460) Fig. 3.5.1 : Two input OR gate

Boolean equation :

The Boolean expression for a two input OR gate is,

$$Y = A + B$$

3.6 The NAND Gate :

SPPU : April 12, April 15

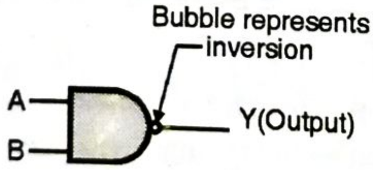
University Questions

Q. 1 Which logic gates are known as universal gates ? Draw their symbols. **(April 12, 2 Marks)**

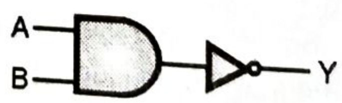
Q. 2 Draw the symbol and truth table for 2 input OR gate and 2 input NAND gate. **(April 15, 2 Marks)**

Logical symbol and truth table :

- The term NAND can be split as NOT-AND which means that the NAND operation can be implemented with the combination of an AND gate and a NOT gate i.e. inverter.
- Thus a NAND gate is equivalent to an AND gate followed by an inverter as shown in Fig. 3.6.1(b).



(a) Logical symbol



(b) Equivalent circuit

Inputs		Output
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(c) Truth table

(B-467) Fig. 3.6.1 : Two input NAND gate

- The symbol of a two input NAND gate is shown in Fig. 3.6.1(a) where a bubble (o) on the output side represents inversion.
- The truth table of a two input NAND gate is shown in Fig. 3.6.1(c), which shows that the output is low (0) if and only if both the inputs are high (1) simultaneously. For all other input combinations the output voltage will be high (1).
- A NAND gate is called as "Universal Gate" because we can construct AND, OR and NOT gates using only NAND gates.

Boolean expression :

- The Boolean expression for the two input NAND gate is,

$$Y = \overline{A \cdot B}$$

Where the line (bar) over the expression $A \cdot B$ represents an inversion.

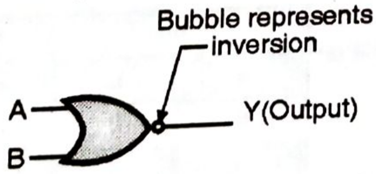
3.7 The NOR Gate :

SPPU : April 12, Oct. 14

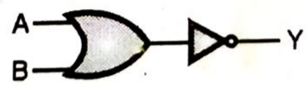
University Questions		
Q. 1	Which logic gates are known as universal gates ? Draw their symbols.	(April 12, 2 Marks)
Q. 2	Draw logic symbol and truth table of NOR gate and NOT gate.	(Oct. 14, 2 Marks)

Logical symbol and truth table :

- This is another universal gate.
- The word NOR can be split as NOT-OR which means that a NOR operation can be implemented with the **combination of an OR gate and a NOT** gate, i.e. inverter. Thus a NOR gate is equivalent to an OR gate followed by an inverter as shown in Fig. 3.7.1(b).
- The symbol of a two input NOR gate is shown in Fig. 3.7.1(a), where a bubble (o) on the output side represents inversion.
- The truth table of a two input NOR gate is shown in Fig. 3.7.1 (c), which shows that "the output of a NOR gate is high (1) if and only if all its inputs are low (0) simultaneously".
- The output of a two input NOR gate is low (0) if any one or all the inputs are at high (1) level.



(a) Logical symbol



(b) Equivalent circuit

Inputs		Output
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

(c) Truth table

(B-471) **Fig. 3.7.1 : A two input NOR gate**

– NOR gate, like NAND gate can be used as a universal gate.

Boolean expression :

– The Boolean expression for the two input NOR gate is given by,

$$Y = \overline{A + B}$$

3.8 EX-OR and EX-NOR Gates : SPPU : April 16, Oct. 16

University Questions

Q. 1 What is meant by derived gate ? Draw the symbol of any two derived gates with their Boolean expression. **(April 16, Oct. 16, 4 Marks)**

– EX-OR and EX-NOR gates are special type of gates. They can be used for applications such as half adder, full adder and subtractors.

– These gates are also called as the derived gates.

3.8.1 The EX-OR Gate : SPPU : Oct. 13, Oct. 15, April 16, Oct. 16, April 17

University Questions

Q. 1 Draw the symbol and write the truth table for 2 input EX-OR gate. **(Oct. 13, 2 Marks)**

Q. 2 Draw logic symbol and write truth table of two input AND and EXOR gates. **(Oct. 15, 2 Marks)**

Q. 3 What is meant by derived gate ? Draw the symbol of any two derived gates with their Boolean expression. **(April 16, Oct. 16, 4 Marks)**

Q. 4 Draw logic diagram of EX-OR gate and explain it's working. **(April 17, 4 Marks)**

Logical symbol and truth table :

– The exclusive-OR gate is abbreviated as EX-OR gate or sometimes as X-OR gate.

– An EX-OR gate having two input terminals and one output terminal is shown in Fig. 3.8.1(a).

(a) Symbol of a two input EX-OR gate

Inputs		Output
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(b) Truth table of a two input EX-OR gate

(B-475) Fig. 3.8.1

– The truth table of a two input EX-OR gate is shown in Fig. 3.8.1(b), which shows that, when both the inputs are at identical logic levels (A = B), the output is low (0) i.e. Y = 0 for A = B = 0 or A = B = 1, and the output is high (1) when A ≠ B.

Boolean expression :

- The Boolean expression for the two input EX-OR gate is,

$$Y = A \oplus B = \bar{A}B + A\bar{B}$$

3.8.2 The EX-NOR Gate :

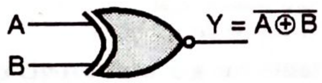
SPPU : April 16, Oct. 16

University Questions

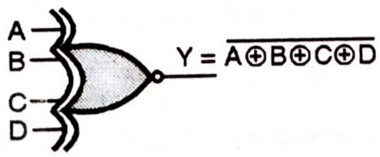
Q. 1 What is meant by derived gate ? Draw the symbol of any two derived gates with their Boolean expression. **(April 16, Oct. 16, 4 Marks)**

Logical symbol and truth table :

- The word EX-NOR is a short form of exclusive-NOR. Exclusive-NOR means NOT-exclusive OR, so EX-NOR gate is equivalent to an EX-OR gate followed by a NOT gate.
- The symbol for a two input EX-NOR gate is as shown in Fig. 3.8.2(a) and its truth table is given in Fig. 3.8.2(c), which shows that the output of an EX-NOR gate is high (1) if both the inputs are identical (A = B) and the output is low (0) if the inputs are not identical (A ≠ B).



(a) Symbol of a two input EX-NOR gate



(b) A four input EX-NOR gate

Inputs		Output
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

(c) Truth table of a two input EX-NOR gate

(B-478) **Fig. 3.8.2**

Boolean expression :

- The Boolean expression for an EX-NOR gate is given by,

$$Y = \overline{A \oplus B} \text{ or } A \odot B = \bar{A}\bar{B} + AB$$

5.1 Boolean (Binary) Algebra :

- Boolean Algebra is used to analyze and simplify the digital (logic) circuits.
- Since it uses only the binary numbers i.e. 0 and 1 it is also called as "Binary Algebra", or "Logical Algebra".
- It was invented by George Boole in the year 1854.
- The rules of Boolean Algebra are different from those of the conventional Algebra in the following manner :
  1. Symbols used in Boolean algebra (usually letters) do not represent numerical values.
  2. Arithmetic operations (addition, subtraction, division etc.) are not performed in Boolean algebra. Also there are no fractions, negative numbers, square, square root, logarithms, imaginary numbers etc.
  3. Third and most important point is Boolean Algebra allows only two possible values ("0" to "1") for any variable.

Rules in boolean algebra :

- There are some rules to be followed while using a Boolean algebra, these are :
  1. Variables used can have only two values. Binary 1 for HIGH and Binary 0 for LOW.
  2. Complement of a variable is represented by a overbar (–). Thus complement of variable B is represented as $\bar{B}$. Thus if B = 0 then $\bar{B}$ = 1 and if B = 1 then $\bar{B}$ = 0.
  3. ORing of the variables is represented by a plus (+) sign between them. For example ORing of A, B, C is represented as A + B + C.
  4. Logical ANDing of the two or more variables is represented by writing a dot between them such as A · B · C · D · E. Sometimes the dot may be omitted like ABCDE.

SPPU : Oct. 12

5.2 Boolean Laws :

University Questions

Q. 1 Give any four basic Boolean postulates and simplify the following expression using Boolean algebra, $A + \bar{A}B = A + B$. **(Oct. 12, 4 Marks)**

- In this section we are going to discuss the following Boolean laws :

1. Commutative law.
2. Associative law.
3. Distributive law.
4. AND law.
5. OR law.
6. INVERSION law.

SPPU : Oct. 15

5.2.1 Commutative Law :

University Questions

Q. 1 Write any four basic Boolean postulates. **(Oct. 15, 2 Marks)**

- Any binary operation which satisfies the following expression is known as commutative operation :

1. $A \cdot B = B \cdot A$
2. $A + B = B + A$

- Thus the commutative law states that we can change the sequence of the variables (inputs) without having any effect on the output of a logic circuit.

5.2.2 Associative Law :

SPPU : Oct. 15

University Questions

Q. 1 Write any four basic Boolean postulates. (Oct. 15, 2 Marks)

- This law states that the order in which the logic operations are performed is not at all important ultimate outcome is the same.

i.e. $(A \cdot B) \cdot C = A \cdot (B \cdot C)$ and $(A + B) + C = A + (B + C)$

5.2.3 Distributive Law :

SPPU : Oct. 15

University Questions

Q. 1 Write any four basic Boolean postulates. (Oct. 15, 2 Marks)

- The distributive law states that, $A \cdot (B + C) = AB + AC$

5.2.4 AND Laws :

SPPU : Oct. 15

University Questions

Q. 1 Write any four basic Boolean postulates. (Oct. 15, 2 Marks)

These laws are related to the AND operation therefore they are called as "AND" laws. The AND laws are as follows :

1. $A \cdot 0 = 0$:

- That means if one input of an AND operation is permanently kept at the LOW (0) level, then the output is zero irrespective of the other variable.

2. $A \cdot 1 = A$:

- That means if one input of an AND operation is HIGH (1) permanently then the output is always equal to the other variable.
- So if $A = 0$, then $Y = 0 \cdot 1 = 0$ i.e. $Y = A$ and if $A = 1$, then $Y = 1 \cdot 1 = 1$ so $Y = A$.

3. $A \cdot A = A$:

- That means if both the inputs in an AND operation have the same value either "0" or "1" then the output will also have the same value as that of the input.

4. $A \cdot \bar{A} = 0$:

- This law states that the result of an "AND" operation on a variable (A) and its complement ($\bar{A}$) is always LOW (0).
- If $A = 0$ then $\bar{A} = 1$ and $Y = 0 \cdot 1 = 0$ whereas if $A = 1$ then $\bar{A} = 0$ and $Y = 1 \cdot 0 = 0$.

5.2.5 OR Laws :

These laws use the OR operation. Therefore they are called as OR laws. The OR laws are as follows :

1. **$A + 0 = A$:**

   – That means if one variable of an "OR" operation is LOW (0) permanently, then the output is always equal to the other variable.

   – $B = 0$ permanently therefore for $A = 0$, $Y = 0 + 0 = 0$ i.e. $Y = A$ and for $A = 1$, $Y = 1 + 0 = 1$ i.e. $Y = A$.

2. **$A + 1 = 1$:**

   – That means if one variable of an "OR" operation is HIGH (1) permanently, then the output is HIGH (1) permanently irrespective of the value of the other variable.

   – Here $B = 1$ permanently, so if $A = 0$ then $Y = 0 + 1 = 1$ and if $A = 1$ then $Y = 1 + 1 = 1$.

   – Thus output remains HIGH (1) always, irrespective of the value of A.

3. **$A + A = A$:**

   – This law states that if both the variables of an OR operation have the same value either "0" or "1" then the output also will be equal to the input i.e. 0 or 1 respectively.

   – For $A = 0$, $Y = 0 + 0 = 0$ i.e. $Y = A$ and for $A = 1$, $Y = 1 + 1 = 1$ i.e. $Y = A$.

4. **$A + \bar{A} = 1$:**

   – This law states that the result of an "OR" operation on a variable and its complement is always 1 (HIGH).

   – If $A = 0$ then $\bar{A} = 1$ and $Y = 0 + 1 = 1$ whereas if $A = 1$ then $\bar{A} = 0$ and $Y = 1 + 0 = 1$.

5.2.6 INVERSION Law :

– This law uses the "NOT" operation. The inversion law states that if a variable is subjected to a double inversion then it will result in the original variable itself i.e.

$$\bar{\bar{A}} = A$$

– Inversion law is being illustrated in Fig. 5.2.1 which shows that if $A = 0$ then $\bar{A} = 1$ and $(\bar{\bar{A}}) = 0 \therefore Y = A$, whereas if $A = 1$ then $\bar{A} = 0$ and $(\bar{\bar{A}}) = 1 \therefore Y = A$.

(B-481) **Fig. 5.2.1 : Illustration of inversion law : $\bar{\bar{A}} = A$**

5.3 De-Morgan's Theorems : SPPU : April 12, April 14, Oct. 14, Oct. 17, April 18

University Questions

Q. 1 State and prove De-Morgan's theorems. (April 12, Oct. 14, Oct. 17, April 18, 4 Marks)

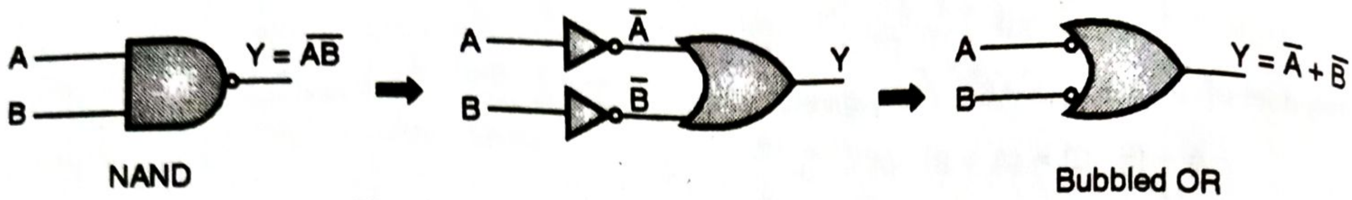
Q. 2 State De-Morgan's theorem. (April 14, 2 Marks)

The two theorems suggested by De-Morgan and which are extremely useful in Boolean algebra are as follows :

Theorem 1 : $\overline{AB} = \bar{A} + \bar{B}$: NAND = Bubbled OR :

- This theorem states that the, complement of a product is equal to addition of the complements.
- This rule is illustrated in Fig. 5.3.1. The Left Hand Side (LHS) of this theorem represents a NAND gate with inputs A and B whereas the Right Hand Side (RHS) of the theorem represents an OR gate with inverted inputs.
- Such an OR gate is called as "Bubbled OR". Thus we can state De-Morgan's first theorem as a NAND operation is equivalent to a bubbled OR operation.

NAND $\equiv$ Bubbled OR



(B-482)Fig. 5.3.1 : Illustration of De-Morgan's first theorem

This theorem can be verified by writing a truth table as shown in Fig. 5.3.2.

A	B	$\overline{AB}$	$\bar{A}$	$\bar{B}$	$\bar{A}+\bar{B}$
0	0	1	1	1	1
0	1	1	1	0	1
1	0	1	0	1	1
1	1	0	0	0	0

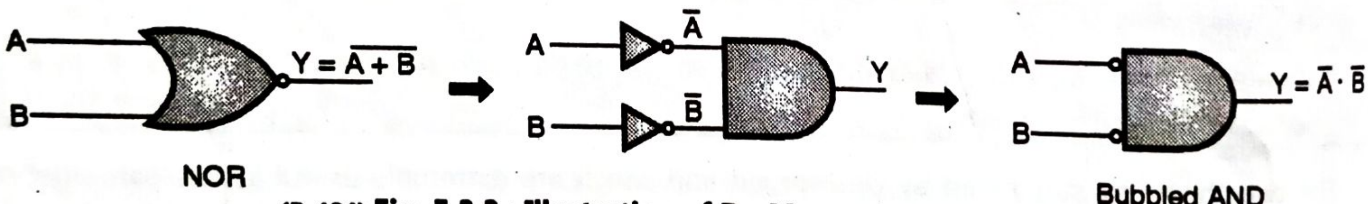
LHS $\overline{AB} = \bar{A} + \bar{B}$ RHS

(C-6138)Fig. 5.3.2 : Verification of the theorem $\overline{A\,B} = \bar{A} + \bar{B}$

Theorem 2 : $\overline{A+B} = \bar{A} \cdot \bar{B}$: NOR = Bubbled AND :

- This theorem is illustrated in Fig. 5.3.3. The LHS of this theorem represents a NOR gate with inputs A and B whereas the RHS represents an AND gate with inverted inputs.
- Such an AND gate is called as "Bubbled AND". Thus we can state De-Morgan's second theorem as a NOR function is equivalent to a bubbled AND function.

NOR ≡ Bubbled AND



(B-484) Fig. 5.3.3 : Illustration of De-Morgan's second theorem

- This theorem can be verified by writing a truth table for both the sides of the theorem statement. This truth table is shown in Fig. 5.3.4, which shows that LHS = RHS.

A	B	$\overline{A+B}$	$\bar{A}$	$\bar{B}$	$\bar{A}\cdot\bar{B}$
0	0	1	1	1	1
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	0

LHS $\overline{A+B} = \bar{A} \cdot \bar{B}$ RHS

(C-6139) Fig. 5.3.4 : Truth table to verify De-Morgan's theorem